

Step 1:
Initialize slope and intercept

$$y = \frac{1}{1 + e^{-(m_1(x_1) + m_2(x_2) + b)}}$$

Because we now have two **independent variables**, we need to extend this equation

$$y = \frac{1}{1 + e^{-(m_1(x_1) + m_2(x_2) + b)}}$$

Let's start with some initial weights:

$$b = -7$$

$$m_1 = 1.2$$

$$m_2 = 0.5$$

Step 2:

Compute probabilities

Compute the probability of Student A

First, let us define $m_1(x_1) + m_2(x_2) + b$ as z

Student	Study Hours (x_1)	Sleep Hours (x_2)	Pass (y)	Predicted ($\hat{y}$)
A	2	5	0	?
B	3	6	0	?
C	4	5	0	?
D	6	6	1	?
E	7	5	1	?

① $z = 1.2(2) + 0.5(5) + (-7) = -2.1$

m_1	m_2	b
1.2	0.5	-7

② $y = \frac{1}{1 + e^{-z}}$

③ $\hat{y} = \frac{1}{1 + e^{-(-2.1)}} = 0.109$

Compute the probability of Student B

First, let us define $m_1(x_1) + m_2(x_2) + b$ as z

Student	Study Hours (x_1)	Sleep Hours (x_2)	Pass (y)	Predicted ($\hat{y}$)
A	2	5	0	0.109
B	3	6	0	?
C	4	5	0	?
D	6	6	1	?
E	7	5	1	?

① $z = 1.2(3) + 0.5(6) + (-7) = -0.4$

m_1	m_2	b
1.2	0.5	-7

② $y = \frac{1}{1 + e^{-z}}$

③ $\hat{y} = \frac{1}{1 + e^{-(-0.4)}} = 0.401$

Compute the probability of Student C

First, let us define $m_1(x_1) + m_2(x_2) + b$ as z

Student	Study Hours (x_1)	Sleep Hours (x_2)	Pass (y)	Predicted ($\hat{y}$)
A	2	5	0	0.109
B	3	6	0	0.401
C	4	5	0	?
D	6	6	1	?
E	7	5	1	?

① $z = 1.2(4) + 0.5(5) + (-7) = 0.3$

② $y = \frac{1}{1 + e^{-z}}$

③ $\hat{y} = \frac{1}{1 + e^{-(0.3)}} = 0.574$

m_1	m_2	b
1.2	0.5	-7

Compute the probability of Student D

First, let us define $m_1(x_1) + m_2(x_2) + b$ as z

Student	Study Hours (x_1)	Sleep Hours (x_2)	Pass (y)	Predicted ($\hat{y}$)
A	2	5	0	0.109
B	3	6	0	0.401
C	4	5	0	0.574
D	6	6	1	?
E	7	5	1	?

① $z = 1.2(6) + 0.5(6) + (-7) = 3.2$

② $y = \frac{1}{1 + e^{-z}}$

③ $\hat{y} = \frac{1}{1 + e^{-(3.2)}} = 0.961$

m_1	m_2	b
1.2	0.5	-7

Compute the probability of Student E

First, let us define $m_1(x_1) + m_2(x_2) + b$ as z

Student	Study Hours (x_1)	Sleep Hours (x_2)	Pass (y)	Predicted ($\hat{y}$)
A	2	5	0	0.109
B	3	6	0	0.401
C	4	5	0	0.574
D	6	6	1	0.961
E	7	5	1	?

① $z = 1.2(7) + 0.5(5) + (-7) = 3.9$

② $y = \frac{1}{1 + e^{-z}}$

③ $\hat{y} = \frac{1}{1 + e^{-(3.9)}} = 0.980$

m_1	m_2	b
1.2	0.5	-7

Prediction Results

Student	Study Hours (x_1)	Sleep Hours (x_2)	Pass (y)	Predicted ($\hat{y}$)
A	2	5	0	0.109
B	3	6	0	0.401
C	4	5	0	0.574
D	6	6	1	0.961
E	7	5	1	0.980

Step 3: Compute Loss

Computing Loss (Binary Cross-Entropy)

$$loss_i = -(\mathbf{y}_i \times \ln(\hat{\mathbf{y}}_i) + (1 - \mathbf{y}_i) \times \ln(1 - \hat{\mathbf{y}}_i))$$

Where:

$\mathbf{y}_i$ is the actual label of each sample

$\hat{\mathbf{y}}_i$ is the **predicted** label of each sample

Compute the loss of Student A

Student	Pass (y)	Predicted ($\hat{y}$)	$loss_i$
A	0	.109	?
B	0	.401	?
C	0	0.574	?
D	1	0.961	?
E	1	0.980	?

$$loss_i = -(\mathbf{y}_i \times \ln(\hat{\mathbf{y}}_i) + (1 - \mathbf{y}_i) \times \ln(1 - \hat{\mathbf{y}}_i))$$

- 1 $loss_i$
- 2 $loss_i = -(0 \times \ln(0.109) + (1 - 0) \times \ln(1 - 0.109))$
- 3 $loss_i = -(\ln(1 - 0.109))$
- 4 $loss_i = -\ln(0.8909) = 0.1155$

Compute the loss of Student B

Student	Pass (y)	Predicted ($\hat{y}$)	$loss_i$
A	0	.109	0.1155
B	0	.401	?
C	0	0.574	?
D	1	0.961	?
E	1	0.980	?

$$loss_i = -(\mathbf{y}_i \times \ln(\hat{\mathbf{y}}_i) + (1 - \mathbf{y}_i) \times \ln(1 - \hat{\mathbf{y}}_i))$$

- 1 $loss_i$
- 2 $loss_i = -(0 \times \ln(0.401) + (1 - 0.401) \times \ln(1 - 0.401))$
- 3 $loss_i = -\ln(1 - 0.401)$
- 4 $loss_i = -\ln(0.5987) = 0.5130$

Compute the loss of Student C

Student	Pass (y)	Predicted ($\hat{y}$)	$loss_i$
A	0	.109	0.1155
B	0	.401	0.5130
C	0	0.574	?
D	1	0.961	?
E	1	0.980	?

$$loss_i = -(\mathbf{y}_i \times \ln(\hat{\mathbf{y}}_i) + (1 - \mathbf{y}_i) \times \ln(1 - \hat{\mathbf{y}}_i))$$

- 1 $loss_i$
- 2 $loss_i = -(0 \times \ln(0.574) + (1 - 0) \times \ln(1 - 0.574))$
- 3 $loss_i = -\ln(1 - 0.574)$
- 4 $loss_i = -\ln(0.4256) = 0.8544$

Compute the loss of Student D

Student	Pass (y)	Predicted ($\hat{y}$)	$loss_i$
A	0	.109	0.1155
B	0	.401	0.5130
C	0	0.574	0.8544
D	1	0.961	?
E	1	0.980	?

$$loss_i = -(y_i \times \ln(\hat{y}_i) + (1 - y_i) \times \ln(1 - \hat{y}_i))$$

- 1 $loss_i$
- 2 $loss_i = -(1 \times \ln(0.961) + (1 - 1) \times \ln(1 - 0.961))$
- 3 $loss_i = -\ln(0.961)$
- 4 $loss_i = 0.0396$

Compute the loss of Student E

Student	Pass (y)	Predicted ($\hat{y}$)	$loss_i$
A	0	.109	0.1155
B	0	.401	0.5130
C	0	0.574	0.8544
D	1	0.961	0.0396
E	1	0.980	?

$$loss_i = -(\mathbf{y}_i \times \ln(\hat{\mathbf{y}}_i) + (1 - \mathbf{y}_i) \times \ln(1 - \hat{\mathbf{y}}_i))$$

- 1 $loss_i$
- 2 $loss_i = -(1 \times \ln(0.980) + (1 - 1) \times \ln(1 - 0.980))$
- 3 $loss_i = -(\ln(0.980) + 0 \times \ln(1 - 0.980))$
- 4 $loss_i = 0.02004$

Computing Loss (Binary Cross-Entropy)

Student	Study Hours (x_1)	Sleep Hours (x_2)	Pass (y)	Predicted ($\hat{y}$)	$loss_i$
A	2	5	0	.109	0.1155
B	3	6	0	.401	0.5130
C	4	5	0	0.574	0.8544
D	6	6	1	0.961	0.03995
E	7	5	1	0.980	0.02004

Compute the Average Loss

$$\textit{Average loss} = \frac{1}{n} \sum_i \textit{loss}_i$$

Where:

$\textit{loss}_i$ is the binary cross entropy loss of each sample
 n is the total number of samples

Compute the Average Loss

Student	Study Hours (x_1)	Sleep Hours (x_2)	Pass (y)	Predicted ($\hat{y}$)	$loss_i$
A	2	5	0	.109	0.1155
B	3	6	0	.401	0.5130
C	4	5	0	0.574	0.8544
D	6	6	1	0.961	0.03995
E	7	5	1	0.980	0.02004

$$\text{Average loss} = \frac{0.1155 + 0.5130 + 0.8544 + 0.03995 + 0.02004}{5} = \frac{1.54}{5} = 0.3086$$

Step 5:

Find the optimal slope and intercept

Find the derivative of loss w.r.t. to intercept *b*

$$\frac{d \text{loss}}{d b} = ?$$

To find the optimal value for the intercept, we need to find the derivative of the *loss* with respect to the intercept *b*

Find the derivative of loss w.r.t. to intercept b

$$\frac{d \text{loss}}{d b} = \boxed{\frac{d \text{loss}}{d \hat{y}}} \times \boxed{\frac{d \hat{y}}{d z}} \times \boxed{\frac{d z}{d b}}$$

The chain-rule states that the derivative of the loss with respect to the intercept b is the product of these derivatives

Find the derivative of loss w.r.t. to intercept b

$$\frac{d \text{loss}}{d b} = \frac{1}{n} \sum_{i=1}^n (\hat{y}_i - y_i)$$

Giving us this equation

Find the derivative of loss w.r.t. to slope m_1

$$\frac{d \text{loss}}{d m_1} = ?$$

To find the optimal value for the slope m_1 , we need to find the derivative of the loss with respect to the intercept m_1

Find the derivative of loss w.r.t. to slope m_1

$$\frac{d \text{loss}}{d m_1} = \boxed{\frac{d \text{loss}}{d \hat{y}}} \times \boxed{\frac{d \hat{y}}{d z}} \times \boxed{\frac{d z}{d m_1}}$$

The chain-rule states that the derivative of the *loss* with respect to the slope m_1 is the product of these derivatives

Find the derivative of loss w.r.t. to slope m_1

$$\frac{d \text{loss}}{d m_1} = \frac{1}{n} \sum_{i=1}^n (\hat{y}_i - y_i) * x_1$$

Giving us this equation

Find the derivative of loss w.r.t. to slope m_2

$$\frac{d \text{loss}}{d m_2} = ?$$

To find the optimal value for the slope m_2 , we need to find the derivative of the loss with respect to the intercept m_2

Find the derivative of loss w.r.t. to slope m_2

$$\frac{d \text{loss}}{d m_2} = \boxed{\frac{d \text{loss}}{d \hat{y}}} \times \boxed{\frac{d \hat{y}}{d z}} \times \boxed{\frac{d z}{d m_2}}$$

The chain-rule states that the derivative of the *loss* with respect to the slope m_2 is the product of these derivatives

Find the derivative of loss w.r.t. to slope m_2

$$\frac{d \text{loss}}{d m_2} = \frac{1}{n} \sum_{i=1}^n (\hat{y}_i - y_i) * x_2$$

Giving us this equation

Derivatives for Gradient Descent

$$\frac{d \text{loss}}{d b} = \frac{1}{n} \sum_{i=1}^n (\hat{y}_i - y_i)$$

$$\frac{d \text{loss}}{d m_1} = \frac{1}{n} \sum_{i=1}^n (\hat{y}_i - y_i) * x_1$$

$$\frac{d \text{loss}}{d m_2} = \frac{1}{n} \sum_{i=1}^n (\hat{y}_i - y_i) * x_2$$

We now have the derivatives for loss b with respect to intercept and loss with respect to the slopes m_1 and m_2

Derivatives for Gradient Descent

$$\frac{d \text{loss}}{d \mathbf{b}} = \frac{1}{n} \sum_{i=1}^n (\hat{y}_i - y_i)$$

$$\frac{d \text{loss}}{d \mathbf{m}_1} = \frac{1}{n} \sum_{i=1}^n (\hat{y}_i - y_i) * x_1$$

$$\frac{d \text{loss}}{d \mathbf{m}_2} = \frac{1}{n} \sum_{i=1}^n (\hat{y}_i - y_i) * x_2$$

We can now use gradient descent to find the optimal value for $\mathbf{b}$, $\mathbf{m}_1$ and $\mathbf{m}_2$

Computing the Gradients

$$\frac{d \text{loss}}{d \mathbf{b}} = \frac{1}{n} \sum_{i=1}^n (\hat{y}_i - y_i)$$

Let us start finding the optimal value for $\mathbf{b}$

Computing the Gradients

$$\frac{d \text{loss}}{d \mathbf{b}} = \frac{1}{n} \sum_{i=1}^n (\hat{y}_i - y_i)$$

According to the formula, to find the optimal value for $\mathbf{b}$, we must get the difference of $\hat{y}_i$ and y_i and get the average.

Computing the Gradients

Student	x_1	x_1	y	$\hat{y}$	$\hat{y} - y$
A	2	5	0	0.1091	0.1091
B	3	6	0	0.4013	0.4013
C	4	5	0	0.5744	0.5744
D	6	6	1	0.9608	-0.0392
E	7	5	1	0.9802	-0.0198

First, we must get the difference of $\hat{y}_i$ and y_i for each row.

Computing the Gradients

Student	x_1	x_1	y	$\hat{y}$	$\hat{y} - y$
A	2	5	0	0.1091	0.1091
B	3	6	0	0.4013	0.4013
C	4	5	0	0.5744	0.5744
D	6	6	1	0.9608	-0.0392
E	7	5	1	0.9802	-0.0198

$$\sum (\hat{y} - y) = 1.0258$$

Get the sum

Computing the Gradients

Student	x_1	x_1	y	$\hat{y}$	$\hat{y} - y$
A	2	5	0	0.1091	0.1091
B	3	6	0	0.4013	0.4013
C	4	5	0	0.5744	0.5744
D	6	6	1	0.9608	-0.0392
E	7	5	1	0.9802	-0.0198

$$\frac{1}{5} \sum_{i=1}^n (\hat{y}_i - y_i) = \frac{1.0258}{5} = 0.205$$

And get the average

Computing the Gradients

Student	x_1	x_1	y	$\hat{y}$	$\hat{y} - y$
A	2	5	0	0.1091	0.1091
B	3	6	0	0.4013	0.4013
C	4	5	0	0.5744	0.5744
D	6	6	1	0.9608	-0.0392
E	7	5	1	0.9802	-0.0198

Which gives us this gradient,

$$\frac{d \text{loss}}{d b} = 0.205$$

Computing the Gradients

$$\frac{d \text{loss}}{d m_1} = \frac{1}{n} \sum_{i=1}^n (\hat{y}_i - y_i) * x_1$$

Let us now find the optimal value for m_1

Computing the Gradients

$$\frac{d \text{loss}}{d m_1} = \frac{1}{n} \sum_{i=1}^n (\hat{y}_i - y_i) * x_1$$

According to the formula, to find the optimal value for b , we must get the difference of $\hat{y}_i$ and y_i , multiply it with x_1 and get the average.

Computing the Gradients

Student	x_1	x_2	y	$\hat{y}$	$\hat{y} - y$	$(\hat{y} - y) * x_1$
A	2	5	0	0.1091	0.1091	0.2182
B	3	6	0	0.4013	0.4013	1.2039
C	4	5	0	0.5744	0.5744	2.2976
D	6	6	1	0.9608	-0.0392	-0.2352
E	7	5	1	0.9802	-0.0198	-0.1386

Get the sum

$$\sum_{i=1}^n (\hat{y}_i - y_i) * x_1 = 3.3459$$

Computing the Gradients

Student	x_1	x_2	y	$\hat{y}$	$\hat{y} - y$	$(\hat{y} - y) * x_1$
A	2	5	0	0.1091	0.1091	0.2182
B	3	6	0	0.4013	0.4013	1.2039
C	4	5	0	0.5744	0.5744	2.2976
D	6	6	1	0.9608	-0.0392	-0.2352
E	7	5	1	0.9802	-0.0198	-0.1386

Get the average

$$\frac{1}{n} \sum_{i=1}^n (\hat{y}_i - y_i) * x_1 = \frac{3.3459}{5} = 0.6692$$

Computing the Gradients

Student	x_1	x_2	y	$\hat{y}$	$\hat{y} - y$	$(\hat{y} - y) * x_1$
A	2	5	0	0.1091	0.1091	0.2182
B	3	6	0	0.4013	0.4013	1.2039
C	4	5	0	0.5744	0.5744	2.2976
D	6	6	1	0.9608	-0.0392	-0.2352
E	7	5	1	0.9802	-0.0198	-0.1386

Which gives us this gradient,

$$\frac{d \text{loss}}{d m_1} = 0.6692$$

Computing the Gradients

$$\frac{d \text{loss}}{d m_2} = \frac{1}{n} \sum_{i=1}^n (\hat{y}_i - y_i) * x_1$$

Let us now find the optimal value for m_2

Computing the Gradients

$$\frac{d \text{loss}}{d m_2} = \frac{1}{n} \sum_{i=1}^n (\hat{y}_i - y_i) * x_2$$

According to the formula, to find the optimal value for b , we must get the difference of $\hat{y}_i$ and y_i , multiply it with x_2 and get the average.

Computing the Gradients

Student	x_1	x_2	y	$\hat{y}$	$\hat{y} - y$	$(\hat{y} - y) * x_1$	$(\hat{y} - y) * x_2$
A	2	5	0	0.1091	0.1091	0.2182	.5455
B	3	6	0	0.4013	0.4013	1.2039	2.4078
C	4	5	0	0.5744	0.5744	2.2976	2.872
D	6	6	1	0.9608	-0.0392	-0.2352	-0.2352
E	7	5	1	0.9802	-0.0198	-0.1386	-0.099

Get the sum

$$\sum (\hat{y} - y) * x_2 = 5.4911$$

Computing the Gradients

Student	x_1	x_2	y	$\hat{y}$	$\hat{y} - y$	$(\hat{y} - y) * x_1$	$(\hat{y} - y) * x_2$
A	2	5	0	0.1091	0.1091	0.2182	.5455
B	3	6	0	0.4013	0.4013	1.2039	2.4078
C	4	5	0	0.5744	0.5744	2.2976	2.872
D	6	6	1	0.9608	-0.0392	-0.2352	-0.2352
E	7	5	1	0.9802	-0.0198	-0.1386	-0.099

Get the average

$$\frac{1}{n} \sum_{i=1}^n (\hat{y}_i - y_i) * x_2 = \frac{5.4911}{5} = 1.09822$$

Computing the Gradients

Student	x_1	x_2	y	$\hat{y}$	$\hat{y} - y$	$(\hat{y} - y) * x_1$	$(\hat{y} - y) * x_2$
A	2	5	0	0.1091	0.1091	0.2182	.5455
B	3	6	0	0.4013	0.4013	1.2039	2.4078
C	4	5	0	0.5744	0.5744	2.2976	2.872
D	6	6	1	0.9608	-0.0392	-0.2352	-0.2352
E	7	5	1	0.9802	-0.0198	-0.1386	-0.099

Which gives us this gradient,

$$\frac{d \text{loss}}{d m_2} = 1.09822$$

Compute Gradients

$$\frac{d \text{loss}}{d \mathbf{b}} = \frac{1}{n} \sum_{i=1}^n (\hat{y}_i - y_i) = \frac{1}{5} (1.0258) = 0.205$$

$$\frac{d \text{loss}}{d \mathbf{m}_1} = \frac{1}{n} \sum_{i=1}^n (\hat{y}_i - y_i) * x_1 = \frac{1}{5} (3.3459) = 0.6692$$

$$\frac{d \text{loss}}{d \mathbf{m}_2} = \frac{1}{n} \sum_{i=1}^n (\hat{y}_i - y_i) * x_2 = \frac{1}{5} (5.4911) = 1.09822$$

Update the Intercept and Slope

$$\frac{d \text{loss}}{d b} = 0.205$$

$$\frac{d \text{loss}}{d m_1} = 0.6692$$

$$\frac{d \text{loss}}{d m_2} = 1.09822$$

m_1	m_2	b
1.2	0.5	-7

Using these gradients, we can now update the **intercept** and **slope**

Update the Intercept and Slope

$$\frac{d \text{ loss}}{d b} = 0.205$$

$$\frac{d \text{ loss}}{d m_1} = 0.6692$$

$$\frac{d \text{ loss}}{d m_2} = 1.09822$$

Step size

$$= \text{learning rate} \times \text{gradient}$$

$$\text{new } b = \text{old } b - \text{Step size}$$

$$\text{new } b = -7 + (0.0205169) = -7.0205169$$

m_1	m_2	b
1.2	0.5	-7
?	?	-7.02

To update b , calculate the Step size and then subtract the **old** b with the Step size.

Update the Intercept and Slope

$$\frac{d \text{ loss}}{d b} = 0.205$$

$$\frac{d \text{ loss}}{d m_1} = 0.6692$$

$$\frac{d \text{ loss}}{d m_2} = 1.09822$$

$$\text{Step size} = \text{learning rate} \times \text{gradient}$$

$$\text{Step size} = 0.1 \times 0.6692 = 0.06692$$

$$\text{new } m_1 = \text{old } b - \text{Step size}$$

$$\text{new } m_1 = -1.2 + (0.06692) = 1.133$$

m_1	m_2	b
1.2	0.5	-7
1.133	?	-7.02

To update m_1 , calculate the Step size and then subtract the **old** m_1 with the Step size.

Update the Intercept and Slope

$$\frac{d \text{ loss}}{d b} = 0.205$$

$$\frac{d \text{ loss}}{d m_1} = 0.6692$$

$$\frac{d \text{ loss}}{d m_2} = 1.09822$$

$$\text{Step size} = \text{learning rate} \times \text{gradient}$$

$$\text{Step size} = 0.1 \times 1.09822 = 0.109822$$

$$\text{new } m_2 = \text{old } b - \text{Step size}$$

$$\text{new } m_2 = 0.5 - (0.109822) = 0.390178$$

m_1	m_2	b
1.2	0.5	-7
1.133	0.390	-7.02

To update m_2 , calculate the Step size and then subtract the **old** m_1 with the Step size.

Step 6:
Repeat Step 2 and 3

Compute new probabilities of Student A

First, let us define $m_1(x_1) + m_2(x_2) + b$ as z

Student	Study Hours (x_1)	Sleep Hours (x_2)	Pass (y)	Predicted ($\hat{y}$)
A	2	5	0	?
B	3	6	0	?
C	4	5	0	?
D	6	6	1	?
E	7	5	1	?

① $z = 1.133(2) + 0.390(5) + (-7.02) = -2.2$

m_1	m_2	b
1.133	0.390	-7.02

② $y = \frac{1}{1 + e^{-z}}$

③ $\hat{y} = \frac{1}{1 + e^{-(-2.2)}} = 0.094$

Compute the probability of Student B

First, let us define $m_1(x_1) + m_2(x_2) + b$ as z

Student4	Study Hours (x_1)	Sleep Hours (x_2)	Pass (y)	Predicted ($\hat{y}$)
A	2	5	0	0.094
B	3	6	0	?
C	4	5	0	?
D	6	6	1	?
E	7	5	1	?

1 $z = 1.133(3) + 0.390(6) + (-7.02) = -0.73$

m_1	m_2	b
1.133	0.390	-7.02

2
$$y = \frac{1}{1 + e^{-z}}$$

3
$$\hat{y} = \frac{1}{1 + e^{-(-0.73)}} = 0.323$$

Compute the probability of Student C

First, let us define $m_1(x_1) + m_2(x_2) + b$ as z

Student	Study Hours (x_1)	Sleep Hours (x_2)	Pass (y)	Predicted ($\hat{y}$)
A	2	5	0	0.094
B	3	6	0	0.323
C	4	5	0	?
D	6	6	1	?
E	7	5	1	?

① $z = 1.133(4) + 0.390(5) + (-7.02) = -0.17$

m_1	m_2	b
1.133	0.39	-7.02

② $y = \frac{1}{1 + e^{-z}}$

③ $\hat{y} = \frac{1}{1 + e^{-(-0.17)}} = 0.457$

Compute the probability of Student D

First, let us define $m_1(x_1) + m_2(x_2) + b$ as z

Student	Study Hours (x_1)	Sleep Hours (x_2)	Pass (y)	Predicted ($\hat{y}$)
A	2	5	0	0.094
B	3	6	0	0.323
C	4	5	0	0.457
D	6	6	1	?
E	7	5	1	?

1 $z = 1.133(6) + 0.390(6) + (-7.02) = 2.5$

m_1	m_2	b
1.133	0.39	-7.02

2 $y = \frac{1}{1 + e^{-z}}$

3 $\hat{y} = \frac{1}{1 + e^{-(2.5)}} = 0.925$

Compute the probability of Student E

First, let us define $m_1(x_1) + m_2(x_2) + b$ as z

Student	Study Hours (x_1)	Sleep Hours (x_2)	Pass (y)	Predicted ($\hat{y}$)
A	2	5	0	0.094
B	3	6	0	0.323
C	4	5	0	0.457
D	6	6	1	0.925
E	7	5	1	?

m_1	m_2	b
1.133	0.39	-7.02

1 $z = 1.133(7) + 0.390(5) + (-7.02) = 3.6$

2 $y = \frac{1}{1 + e^{-z}}$

3 $\hat{y} = \frac{1}{1 + e^{-(3.6)}} = 0.974$

New Probabilities

Student	Study Hours (x_1)	Sleep Hours (x_2)	Pass (y)	Predicted ($\hat{y}$)
A	2	5	0	0.094
B	3	6	0	0.323
C	4	5	0	0.457
D	6	6	1	0.925
E	7	5	1	0.974

Compute the loss of Student A

Student	Pass (y)	Predicted ($\hat{y}$)	$loss_i$
A	0	0.094	?
B	0	0.323	?
C	0	0.457	?
D	1	0.925	?
E	1	0.974	?

$$loss_i = -(\mathbf{y}_i \times \ln(\hat{\mathbf{y}}_i) + (1 - \mathbf{y}_i) \times \ln(1 - \hat{\mathbf{y}}_i))$$

- 1 $loss_i$
- 2 $loss_i = -(0 \times \ln(0.094) + (1 - 0) \times \ln(1 - 0.094))$
- 3 $loss_i = -(\ln(1 - 0.094))$
- 4 $loss_i = -\ln(0.906) = 0.099$

Compute the loss of Student B

Student	Pass (y)	Predicted ($\hat{y}$)	$loss_i$
A	0	0.094	0.098
B	0	0.323	?
C	0	0.457	?
D	1	0.925	?
E	1	0.974	?

$$loss_i = -(\mathbf{y}_i \times \ln(\hat{\mathbf{y}}_i) + (1 - \mathbf{y}_i) \times \ln(1 - \hat{\mathbf{y}}_i))$$

- 1 $loss_i$
- 2 $loss_i = -(0 \times \ln(0.323) + (1 - 0.323) \times \ln(1 - 0.323))$
- 3 $loss_i = -\ln(1 - 0.323)$
- 4 $loss_i = -\ln(0.677) = 0.390$

Compute the loss of Student C

Student	Pass (y)	Predicted ($\hat{y}$)	$loss_i$
A	0	0.094	0.098
B	0	0.323	0.390
C	0	0.457	?
D	1	0.925	?
E	1	0.974	?

$$loss_i = -(\mathbf{y}_i \times \ln(\hat{\mathbf{y}}_i) + (1 - \mathbf{y}_i) \times \ln(1 - \hat{\mathbf{y}}_i))$$

- 1 $loss_i$
- 2 $loss_i = -(0 \times \ln(0.457) + (1 - 0) \times \ln(1 - 0.457))$
- 3 $loss_i = -\ln(1 - 0.457)$
- 4 $loss_i = -\ln(0.543) = 0.611$

Compute the loss of Student D

Student	Pass (y)	Predicted ($\hat{y}$)	$loss_i$
A	0	0.094	0.098
B	0	0.323	0.390
C	0	0.457	0.611
D	1	0.925	?
E	1	0.974	?

$$loss_i = -(\mathbf{y}_i \times \ln(\hat{\mathbf{y}}_i) + (1 - \mathbf{y}_i) \times \ln(1 - \hat{\mathbf{y}}_i))$$

- 1 $loss_i$
- 2 $loss_i = -(1 \times \ln(0.925) + (1 - 1) \times \ln(1 - 0.925))$
- 3 $loss_i = -\ln(0.925)$
- 4 $loss_i = 0.077$

Compute the loss of Student E

Student	Pass (y)	Predicted ($\hat{y}$)	$loss_i$
A	0	0.094	0.098
B	0	0.323	0.390
C	0	0.457	0.611
D	1	0.925	0.077
E	1	0.974	?

$$loss_i = -(\mathbf{y}_i \times \ln(\hat{\mathbf{y}}_i) + (1 - \mathbf{y}_i) \times \ln(1 - \hat{\mathbf{y}}_i))$$

- 1 $loss_i$
- 2 $loss_i = -(1 \times \ln(0.974) + (1 - 1) \times \ln(1 - 0.974))$
- 3 $loss_i = -(\ln(0.974) + 0 \times \ln(1 - 0.974))$
- 4 $loss_i = 0.026$

Computing Loss (Binary Cross-Entropy)

Student	Study Hours (x_1)	Sleep Hours (x_2)	Pass (y)	Predicted ($\hat{y}$)	$loss_i$
A	2	5	0	0.094	0.098
B	3	6	0	0.323	0.390
C	4	5	0	0.457	0.611
D	6	6	1	0.925	0.077
E	7	5	1	0.974	0.026

Compute the Average Loss

Student	Study Hours (x_1)	Sleep Hours (x_2)	Pass (y)	Predicted ($\hat{y}$)	$loss_i$
A	2	5	0	0.094	0.098
B	3	6	0	0.323	0.390
C	4	5	0	0.457	0.611
D	6	6	1	0.925	0.077
E	7	5	1	0.974	0.026

$$\text{Average loss} = \frac{0.098 + 0.390 + 0.611 + 0.077 + 0.026}{5} = \frac{1.202}{5} = 0.2404$$